

# DRAFT: Formalism of 4D Filamentary Dynamics: The Master Lagrangian and Universal Indicatrix Mapping

Nathan McKnight  
*Independent Researcher*  
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This paper outlines a geometric framework where particle worldlines are modeled as four-dimensional (4D) superhelical filaments. We propose that physical observables arise from the projection of these 4D paths into a 3D intersection surface. We define a Master Lagrangian that integrates filament coiling energy with lattice dynamics and a Universal Indicatrix (UI) mapping function for 3D projection. By utilizing a zero-parameter economy, we derive mass scaling laws and relativistic thresholds directly from the geometry of a 24-cell lattice, eliminating the need for manually tuned coupling constants.

## I. INTRODUCTION

Modern physics often relies on point-particle abstractions that require the manual insertion of independent constants. This framework proposes an alternative: matter as persistent 4D filamentary worldlines. These filaments exhibit recursive superhelical coiling, a geometric structure that models internal degrees of freedom.

In this model, the 3D "present" is a propagation surface that intersects the 4D lattice. Mass and inertia are recontextualized as the projective resistance encountered by this surface as it moves along the time-flow vector through varying filament densities. By adopting this neutral geometric metaphor, we can derive physical properties—such as mass, spin, and charge—as integrated invariants of the filament's coiling history and its topological charge ( $Q$ ).

## II. THE MASTER LAGRANGIAN

The dynamics of these recursive filamentary structures are governed by an action that integrates internal coiling energy, inter-filament interactions, and torsion.

### A. Dynamics of Superhelical Filaments

The Lagrangian for the system replaces manual coupling constants with geometric tensors derived from the lattice architecture:

$$L_{SAT} = \sum_i \sum_o \left[ \frac{1}{2} \mu_i \left| \frac{d\lambda}{dH} \right|^2 + \sum_j \kappa F_{\text{braid}}(H_i, H_j) + \alpha \left( \frac{d\lambda}{dH} \cdot T \right)^2 \right] \quad (1)$$

where  $\lambda$  is the filament parameter,  $H$  is the 4D lattice coordinate,  $T$  is the torsion vector, and  $F_{\text{braid}}$  represents the interaction energy between interwoven filaments. This formulation ensures that internal coiling energy is balanced against the constraints of the 4D lattice.

## B. Universal Indicatrix Mapping

To project the 4D filament paths ( $x_0^\nu$ ) into observable 3D trajectories ( $y^\mu$ ), we define the Universal Indicatrix (UI):

$$y^\mu(\lambda) = r(\lambda) R_\nu^\mu(\lambda) x_0^\nu \quad (2)$$

This mapping function incorporates geometric deformation corrections required to maintain structural integrity during the  $4D \rightarrow 3D$  projection. The vacuum state is defined by a ground-state geometric effort of exactly 0.5 units:

$$L_{UI} = \frac{1}{2} \dot{r}^2 + \frac{1}{2} r^2 \Omega_{\mu\nu} \Omega^{\mu\nu} = 0.5 \quad (3)$$

This baseline establishes the "true vacuum" as a stable geometric state.

## III. DISCUSSION

### A. Zero-Parameter Economy

This framework utilizes a scale-free "Zero-Parameter Economy." Rather than relying on arbitrary constants, all physical values are derived from a single dimensionful anchor—the universal mass anchor  $m_0 \approx 1.0073 \times 10^{-27}$  kg—and the specific geometry of the 24-cell lattice.

### B. Anchoring and Scaling

Anchoring is the procedural fixing of the lattice scale, which then propagates through all physical sectors. Mass scaling follows strict geometric laws:

1. **Baryon Scaling:**  $m_B = 2B^Q \cdot m_0$ , where  $B \approx 0.2387$  rad is the projection constant.
2. **Lepton Scaling:**  $m_L = Q \cdot m_0 \cdot B_4$ , utilizing a fractional gear ( $B/4$ ) derived from lattice packing.
3. **High-Density Corrections:** For complex knots (e.g., Helium-3), a braid-smoothing metric is applied to account for non-linear interference.

### C. Implementation via Universal Indicatrix

The UI serves as the primary tool for scientists to test the framework. By setting the UI parameters to the vacuum baseline, one can insert filament trajectories and observe how they project into 3D space. This projection inherently predicts relativistic thresholds, such as the critical velocity  $v_{crit} \approx 0.2387c$ , beyond which matter precesses out of the observable 3D slice at an angle of  $\theta_{obs} \approx 14.1^\circ$  (0.246 rad).

### IV. EMPIRICAL MATCHING

Table I demonstrates the agreement between derived geometric constants and experimental data.

TABLE I. Comparison of Predicted vs. Measured Values.

| Constant                                | Predicted Value          | Source/Match |
|-----------------------------------------|--------------------------|--------------|
| Projection Constant ( $B$ )             | $\approx 0.2387$ rad     |              |
| Obscuration Angle ( $\theta_{obs}$ )    | $14.1^\circ$ (0.246 rad) |              |
| Critical Velocity ( $v_{crit}$ )        | $\approx 0.2387c$        |              |
| Lattice Mode Density ( $\rho_{embed}$ ) | $\approx 10^{-19}$       |              |
| Geometric Effort (Vacuum)               | 0.5 units                |              |

### V. CONCLUSION

The Master Lagrangian and Universal Indicatrix provide a mathematically closed system for deriving 3D physics from 4D mechanical rotations. Falsification remains possible through the detection of discrete 0.246 rad phase snaps in pulsar timing or the verification of global curvature corrections ( $1/R^2$ ) in atomic energy ladders.

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